25901912 Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMAT201 - BUILDING BLOCKS FOR HIGHER MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

{Short Answer Questions}

Answer any **5** questions

Each question carries **2** marks

1. Explain the method of proof by contradiction. [K] / [CO1]

2. Find the number of relations from [U] / [CO2]

   $A = \{a, b, c\} \quad \text{to} \quad B = \{1, 2\}.$

3. Check if { -3, -1, 1, 3} and {-2, 0, 2} form a partition of the set {-3, -2, -1, 0, 1, 2, 3}. [U] / [CO3]

4. Give an example of an infinite lattice which has neither a least nor a greatest element [U] / [CO3]

5. Find the unit tangent vector of the plane curve $\mathbf{r}(t) = \cos^3 t\mathbf{i} + \sin^3 t\mathbf{j}$. [A] / [CO4]

6. Write a Vector equation for the unit circle in the XY – Plane centred at Origin. [A] / [CO4]

7. Define line integral of f over a curve C. [K] / [CO5]

8. Test whether the vector field $\mathbf{F} = (z + y)\mathbf{i} + z\mathbf{j} + (y + x)\mathbf{k}$ is conservative. [U] / [CO5]

[2x5 = 10]

Part B

{Short Essay Questions}

Answer any **5** questions

Each question carries **6** marks

9. (i) What do you mean by "begging the question"? [A] / [CO1]
   (ii) Is the following argument correct? It supposedly shows that n is an even integer whenever n^2 is an even integer.
   Suppose that n^2 is even. Then $n^2 = 2k$ for some integer k. Let $n = 2l$ for some integer l. This shows that n is even.

10. (i) What is meant by a direct proof? [A] / [CO1]
    (ii) Give a direct proof to show that the sum of two even integers is even.

11. Write all the elements in $S \times S$ in lexicographic order based on the "less than" relation where $S = \{1, 2, 3, 4\}$. [U] / [CO3]

12. Determine all minimal, maximal, least, and greatest elements of the poset ({1, 2, 3, 4, 6}, |) [U] / [CO3]

13. Find the arc length parameter $s(t)$ along the curve $\mathbf{r}(t) = (4\cos t)\mathbf{i} + (4\sin t)\mathbf{j} + 3t\mathbf{k}$ from the point where $t = 0$ by evaluating the integral $s = \int_0^t |\mathbf{v}(\tau)|\, d\tau$. [A] / [CO4]
    Use this arc length parameter to find the length of the curve for $0 \le t \le \dfrac{\pi}{2}$.

14. Find unit tangent vector **T** and curvature for the helix $r(t) = (3\cos t)\mathbf{i} + (3\sin t)\mathbf{j} + 4t\ \mathbf{k}$. [A] / [CO4]

15. Evaluate $\int_{(1,1,1)}^{(2,3,-1)} y\, dx + x\, dy + 4\, dz$ [A] / [CO5]

16. A fluid's velocity field is $\vec{F} = (x - y)\mathbf{i} + x\mathbf{j}$. Find the circulation along the curve $\overrightarrow{r(t)} = cost\mathbf{i} + sint\mathbf{k}, 0 \le\ t \le 2\pi$. [An] / [CO5]

[6x5 = 30]

Part C

{Essay Questions}

Answer any **3** questions

Each question carries **10** marks

17. Determine whether the relations represented by the matrices are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive. a) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, b) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, c) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, d) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. [An] / [CO2]

18. Prove that every non-empty set of natural numbers, ordered by the usual relation $\leq$ is well-ordered. [U] / [CO3]

19. Let R be the set of all real numbers. Define a relation R by ${}_xR_y$ if and only if $|x - y| > 0$. Check all properties of an equivalance relation and determine whether R is an equivalence relation. [K] / [CO3]

20. Find the derivative of the function at P_0 in the direction of u. [U] / [CO4]

a) $f(x, y, z) = x^2 + 2y^2 - 3z^2$, $P_0(1, 1, 1)$, $\mathbf{u} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

b) $g(x, y, z) = 3e^x \cos(yz)$, $P_0(0, 0, 0)$, $\mathbf{u} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$

21. Evaluate the line integral of a vector field [A] / [CO5]
$\mathbf{F}(x, y, z) = (y + z)\,\mathbf{i} + (z + x)\,\mathbf{j} + (x + y)\,\mathbf{k}$ along the path
$C_1 \cup C_2$ consisting of the line segment from (0, 0, 0) to (1, 1, 0) followed by the segment from (1, 1, 0) to (1, 1, 1).

22. Find the divergence of each of the following vector field and interpret its physical meaning:. [A] / [CO5]
(a) Expansion: $\mathbf{F}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$.
(b) Compression: $\mathbf{F}(x, y, z) = -x\hat{i} - y\hat{j} - z\hat{k}$.
(c) Rotation about z-axis: $\mathbf{F}(x, y, z) = -y\hat{i} + x\hat{j}$.

[10x3 = 30]